Comp. Phys. Lab for Statistical Mechanics
Fall Quarter 2026
Physics 112L



Michael Mulhearn
mulhearn@physics.ucdavis.edu
Physics 317

Lecture: Tuesday 2:10-3:00 PM in Roessler 55

Textbooks: Thermal Physics by Daniel V. Schroeder (any volume)
(Recommended for reference, or any other Intro. Stat Mech text.)
Lecture notes on Scientific Python (link on course site)

Course TA: TBD

Office Hours:

Mulhearn:	Mondays 10:30 AM-12:10 PM (weeks HW is due) Thursdays 1:30 PM-2:50 PM (otherwise)
TA:	TBD
(H-bar)	Wednesdays 3:00 PM - 4:00 PM in Physics 285

Quizzes: Short in-class quizzes, in the week after each assignment is due.
Final Project: Due TBD

Course Description:
Applications of computational physics to problems from Statistical Mechanics. Application of numerical techniques, and comparison to analytic solutions (where they exist).

Course Structure:
The course runs on a two-week cycle. In the first week of each cycle, the previous assignment is due and the next assignment is described in lecture. In the second week, the lecture period begins with a short in-class quiz on the assignment you just submitted, and the remaining time is available for questions and help with the current assignment. The last part of the

quarter is devoted to the final project.

System of Units:
To receive any credit, your solutions must adopt the system of units as presented in lecture and in the write-up for each assignment. The system of units must be used throughout your solution. You may not e.g. do the calculation in SI and then convert to our system of units later. If you see physical constants like Boltzmann's constant or Avogadro's number appearing in your code, that is likely a misunderstanding.

Use of AI and Other External Resources:
You must adhere to the UC Davis Code of Academic Conduct, available online at the OSSJA website. In particular, you may not take credit for any work not created by you. **Use of generative AI tools in this course is forbidden unless explicitly allowed below.** Likewise, you may not direct search engines, homework help websites, or other external resources toward the specific problems that are assigned to you.

The following uses of AI are allowed:

- **Learning Python.** You may use AI to learn or review general features of Python and its libraries (e.g. how `numpy` arrays or `matplotlib` plots work). You may not give an AI any part of an assignment's text or your own assignment code, or ask it to write, debug, or explain code for an assignment.
- **Final project.** There are no restrictions on AI use for the final project.

If you use AI (or other external resources) on an assignment in a way not allowed above, you must disclose it. The assignment will receive no credit, and nothing further will happen: disclosure is treated as a missed assignment, not as misconduct. If you do not disclose it, it will be referred to OSSJA, and if you are found responsible, you will fail the course.

You are allowed to collaborate with other students in the course, including discussing specific problems in the homework, unless you have reason to believe the students are using unauthorized methods. The solution that you provide must be your own: each and every line. In general, you should be able to explain why each line in your code is there. If you do not understand something in your solution because you relied on trial and error, then you should state that in the comments, including explaining any other things that you tried. For example:

```
# I don't understand why this factor of two is needed
# I have included it because it gives the correct result:
```

Quizzes:
Each assignment is followed by a short, closed-notes quiz in class. The quiz is not something to study for in the usual sense: its purpose is for you to show that you wrote, and understand, the code you submitted. Most questions will be about *your own* submitted notebook:

for example, what would happen to your result if a particular line or parameter in your code were changed, or why a particular line is needed. These questions are chosen from your own submitted code, with help from the tools described under Homework below. Some questions will be common to everyone and cover the physics and numerical methods of the assignment. If you wrote your own code and understand it, you should be well prepared. Failing the quiz means failing the corresponding assignment.

Oral exam option. If you pass all submitted assignments but your course grade is failing due to quiz performance, you may request an oral exam to demonstrate that the assignment work is your own. If you do, your course grade will be at least a C. If not, your grade stands. Requests must be made within N days of grades being posted.

Course Philosophy:
Computational physics is essential to nearly every career related to physics. Unfortunately, the educational pipeline, since grade school, does not adequately prepare students with programming skills, and so a central feature of computational physics courses is a wide distribution of student programming abilities. There are students that are better programmers than most physics professors, students that have little practice outside of the required computational courses, and everything in between.

Fortunately, in this course we are concerned with **computational physics** and not just **computer programming**. You do not need to be an expert computer programmer to succeed in this course. You do need to gain enough programming competency to handle the fascinating physics problems we will tackle. **The primary objective of this course is to make you confident in using programming independently as an essential and effective tool for solving physics problems.** AI tools are powerful, and you will be free to use them on the final project, but that is only valuable once you can program independently and judge whether what the AI produces is correct. That independence is what the homework is designed to build.

This is a one unit course, and I will do my best to keep the workload appropriate. However, given the wide variation in student programming abilities, inevitably some students will find the homework more time-consuming than others. If you do find yourself working longer on the assignments, please understand that this is time well-spent: gaining more proficiency in computing is almost certainly one of the best possible uses of your time at this stage in your career. There are a vanishingly small number of physics-related career options that do not require programming ability.

Homework:
There will be four homework assignments featuring problems in computational physics. Each assignment is graded pass/fail, and each is followed by an in-class quiz as described above.

Before submitting, restart your notebook and run every cell in order (Kernel → Restart &

Run All), so that the output you submit is the output of the code you submit. Use the function names and the reporting of results required by each assignment: the automated tests described below depend on them.

Do not include any personal information in your submitted assignments, including your name, student ID, or email address, in the notebook, its file name, or its comments. The submission system already records who submitted each assignment.

How your submissions are processed. Each assignment comes with a short checker program, which you run in the last cell of your notebook before submitting. It reports anything that is missing and whether your results are correct to within the expected precision. Submit a notebook that passes the checker; if yours does not pass, explain why in a text cell at the end of the notebook. After you submit, your notebook is processed by automated tools, including AI models that run only on a department computer under the instructor's control: for example, to run additional tests, to summarize your results, and to help prepare the questions for your individual quiz. **Your work is never sent to any external or cloud-based AI service or other third party.** All grading and quiz decisions are made by the instructor and the TA, not by these tools.

Final Project:
In lieu of a final exam, you will complete a final project. I will provide a set of open-ended, deliberately ambiguous questions. You will choose one, build a numerical model that illuminates it, and present your results. You should be prepared to explain and defend your modeling choices. AI use is unrestricted on the final project. Further details will be provided in lecture.

Grades:
Your final grade will be determined from the homework, the quizzes, and the final project. You will be allowed to submit one homework assignment late, but you must complete every assignment and your final project to complete the course. You may also miss one quiz without penalty (for example, the quiz for an assignment you submit late).

Contacting the Instructor:
I have posted my email, but that is mainly so that you can contact me after the course has ended. During the course, the most reliable way to reach me is by talking to me after lecture or coming to office hours. If you must reach me remotely, please use the course website instead of email.